

Here is the complete, fresh, step-by-step setup guide for anyone cloning your repository. You can copy this directly into your `README.md`.

```
# Smart Patient-Doctor Telemedicine Platform
```

```
A full-stack telemedicine platform built with a React + Vite frontend and a FastAPI + PostgreSQL backend. It features an AI-powered symptom checker (RAG + safety layers), zero-friction Jitsi video consultations, and digital PDF prescription generation.
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## 📋 Prerequisites
```

```
Before you begin, ensure you have the following installed on your system:
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```
* Python 3.11+
```

```
* Node.js 18+
```

```
* PostgreSQL 14+ (Must support the pgvector extension)
```

```
* OpenAI API Key (Get a free key from [openai.com])
```

2. Backend Setup (FastAPI)

Open your terminal and navigate into the `backend` directory.

```
cd backend
```

```
# Create and activate a Python virtual environment
```

```
python -m venv .venv
```

```
source .venv/bin/activate # On Windows use: .venv\Scripts\activate
```

```
# Install required Python packages
```

```
pip install -r requirements.txt
```

```
# Create your local environment configuration file
cp .env.example .env
```

Edit your `backend/.env` file to match your local configuration:

```
DATABASE_URL=postgresql+psycopg://postgres:postgres@localhost:5432/te
lmedicine
SECRET_KEY=your_super_secret_random_string_here
GROQ_API_KEY=your_actual_groq_api_key_here
```

3. Database Migrations, Seeding & AI Knowledge Base

Ingestion

Run these commands sequentially from inside the `backend/` directory with your virtual environment active:

```
# 1. Apply database migrations (creates all tables, enums, indexes,
and meeting_link columns)
alembic upgrade head

# 2. Seed core data (creates Admin account, expanded doctors with
shifts, and sample patients)
python -m app.database.seed

# 3. Ingest medical knowledge base chunks (Crucial for the AI Symptom
Checker / RAG vector retrieval)
python -m app.scripts.ingest_knowledge
```

4. Frontend Setup (React / Vite)

Open a **second, separate terminal window**, navigate into the `frontend` directory, and install dependencies:

```
cd frontend
```

```
# Install Node dependencies  
npm install
```

(Optional: If your frontend needs an environment file pointing to the backend API, ensure `frontend/.env` contains

```
VITE_API_BASE_URL=http://localhost:8000/api/v1 )
```

5. Running the Application

You will need to keep both terminals running simultaneously.

Terminal 1 — Start the Backend Server

```
cd backend  
source .venv/bin/activate  
uvicorn app.main:app --reload --port 8000
```

- **API Base URL:** `http://localhost:8000/api/v1`
- **Interactive API Docs (Swagger):** `http://localhost:8000/docs`

Terminal 2 — Start the Frontend Development Server

```
cd frontend  
npm run dev
```

- **Frontend App URL:** `http://localhost:5173`

6. Testing the End-to-End Flow

Use the pre-seeded admin account to bootstrap your platform testing:

- **Admin Email:** `admin@telemedicine.local`
- **Admin Password:** `ChangeMe123!`

Suggested Manual Test Sequence:

1. **Admin Approval:** Log in with the admin credentials above to approve the pre-seeded doctors so they appear in public search.
2. **Patient Booking:** Register a new patient account (or log in as `patient.one@telemedicine.local` / `ChangeMe123!`), search for an approved doctor, pick an available slot, and book an appointment.
3. **AI Symptom Checker:** Test the AI assistant on the patient dashboard (e.g., typing symptoms like a rash or chest pain to verify triage, specialty mapping, and safety rules).
4. **Video Call Flow:**
 - Open two browser windows side-by-side (Window 1: Regular window logged in as **Patient**; Window 2: Incognito window logged in as **Doctor**).
 - In Window 2 (Doctor), go to today's appointments, confirm the booking, mark it waiting, and click **Start Consultation**.
 - Watch Window 1 (Patient) auto-refresh via polling within 10 seconds to reveal the  **Join Video Call** button.
 - Click the video button in both windows to instantly enter the secure, zero-login Jitsi room (`meet.ffmuc.net`).
5. **Prescriptions:** During or after the consultation, the doctor can write a digital multi-medicine prescription, and the patient can instantly view and download it as a formatted PDF.